

QUANTUM MECHANICS

classical concepts like laws of newton, laws of thermodynamics, classical mechanics aren't valid in atomic dimension of electron, proton, neutron, nucleus.

In microscopic world of atoms, energy & momentum didn't undergo change in continuous manner from one state to another as expected under classical laws. Energy & momentum had a discrete value in a particular state of atom. That's why, to include new concepts newtonian laws were modified which give rise to quantum mechanics.

→ INADEQUACY OF CLASSICAL MECHANICS

→ it fails to explain :-

1. Phenomena related to motion of microscopic object
2. Stability of atom: electron moves around nucleus as earth moves about sun but unlike planets energy of e^- would decrease as it emits energy in form of e.m. waves. Thus, e^- comes close to nucleus & would lead to atom collapsed.
3. Variation of specific heat with temp of metal
4. Dispersion of light
5. Black body radiation
6. Spin of e^- (Pauli exclusion Principle)
7. Superconductivity (zero Resistance, ∞ conductance)
8. Hydrogen spectrum

HYDROGEN SPECTRUM

$$\frac{1}{\lambda} = R Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$R \rightarrow$ Rydberg's constant
 $n_1 < n_2$

experimentally classical mechanics observed H spectrum consists of discrete lines.

Acc to classical theory, excited H atom emits electromagnetic radiations of all wavelengths. continuously emitted radiations have certain wavelength. This difficulty was later resolved by Bohr content.

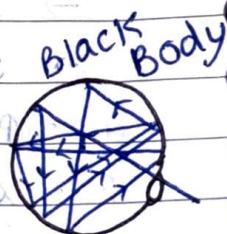
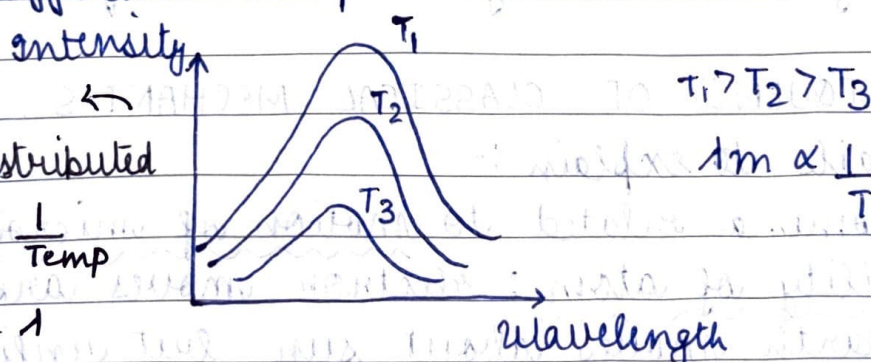
BLACK BODY RADIATION

variation of intensity of radiation with λ at different temp.

1. Energy isn't uniformly distributed

2. Intensity $\propto \frac{1}{\text{Temp}}$

3. Intensity $\propto \lambda$



Black body is a type of body which absorbs all the heat radiations of all frequencies incident on it. Such body is represented by a hollow container with a very small hole in the wall. When it's heated, emits radiation of each type of wavelength which are independent of nature of substance. As it absorbs, it emits too.

A perfectly black body never exists.

- For all wavelengths, an increase in temp cause increase in energy emission
- The area under each curve represents total energy for complete spectrum at particular temp.

→ Two fundamental laws of black body radiation on the basis of classical theory, attempt was made to explain qualitatively & quantitatively

- STEPHAN BOLTZMANN LAW

This was established experimentally by Stephan. Acc. to this law, intensity of total radiation E is proportional to fourth power of temp T

$$E = \sigma T^4$$

Stefan's constant

$$\sigma = 5.672 \times 10^{-8}$$

It was thermodynamically derived by Boltzmann

- WEIN'S DISPLACEMENT LAW

Acc to it, wavelength of max spectral distribution (λ_{max}) is inversely prop to temp T

$$\lambda_{max} \times T = \text{Constant}$$

$$\lambda_m \propto \frac{1}{T}$$

Also, $E \propto T^5$

Based on classical electromagnetic theory, assuming oscillators from which radiation come are of molecular size deduced:-

$$E = \frac{a}{15} \left[e^{-b/1T} \right]$$

$E \rightarrow$ energy emitted b/t $1.8 \text{ } 1^\circ\text{C}$
 $a \text{ \& } b \rightarrow$ constants

→ DRAWBACK OF WEIN'S

It was only suitable for curves in low wavelength region.

→ RAYLIEGH JEANS LAW

Applied classical law of equipartition of energy; each vibrational mode has average energy $\bar{e} = KT$ to the oscillators of black body.

$$\bar{e} = KT$$

It was seen no. of modes of oscillation dn in range λ to $\lambda + d\lambda$ is equal to $dn \cdot \bar{e}$ per unit volume is $\frac{8\pi d\lambda}{\lambda^4}$

$$\star \quad dn = \frac{8\pi d\lambda}{\lambda^4}$$

Also energy density in this range is $dn \cdot \bar{e}$ which is equal to $E_\lambda d\lambda$

$$\begin{aligned} E_\lambda d\lambda &= dn \cdot \bar{e} \\ \Rightarrow E_\lambda d\lambda &= \left(\frac{8\pi d\lambda}{\lambda^4} \right) (KT) \end{aligned}$$

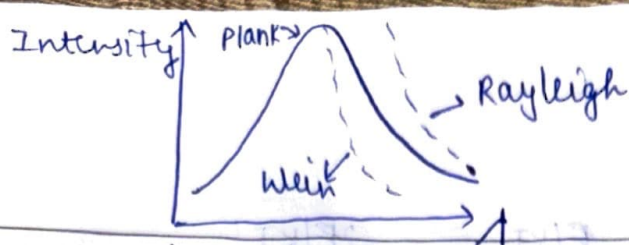
$$\Rightarrow E_\lambda = \frac{8\pi KT}{\lambda^4}$$

$$E_\lambda \propto \frac{1}{\lambda^4} \Rightarrow E \uparrow \lambda \downarrow$$

→ DRAWBACK

Completely fails at low wavelength but fits high wavelength

leads to ultraviolet catastrophe



→ PLANCK'S RADIATION LAW

Heinrich Hertz, Rayleigh's laws, gave his law on these assumptions:-

- A black body radiator contains simple harmonic oscillators of molecular dimension which can vibrate with all possible freq. Equipartition isn't applicable but assumed oscillator of the black body have a discrete energy of

$$e = nh\nu$$

∴ it would lead to Rayleigh Jeans

- oscillator don't radiate / absorb continuously but exchange energy with surroundings in discrete values - $0, h\nu, 2h\nu, 3h\nu \dots nh\nu$

Suppose $N \rightarrow$ total no of planck's oscillator's
 $E \rightarrow$ total energy

$$\bar{e} = \frac{E}{N} \quad \left. \vphantom{\frac{E}{N}} \right\} \text{energy per oscillator}$$

If $N_0, N_1, N_2 \dots N_n$ are oscillators having energies $0, e, 2e, 3e$ then

$$N = N_0 + N_1 + N_2 \dots N_n$$

$$E = 0 + N_1 e + 2N_2 e + 3e N_3 \dots n e N_n$$

From maxwell formula

$$N_n = N_0 e^{-\frac{ne}{KT}}$$

$$\underline{N_n = N_0 e^{-\frac{ne}{KT}}}$$

$$N_1 = N_0 e^{-\frac{e}{KT}}$$

$$N_2 = N_0 e^{-\frac{2e}{KT}}$$

$$\Rightarrow N = N_0 + N_0 e^{-E/KT} + N_0 e^{-2E/KT} \dots N_0 e^{-\infty E/KT}$$

$$\Rightarrow \underline{N = \frac{N_0}{1 - e^{-E/KT}}}$$

$$\Rightarrow E = 0 + E N_0 e^{-E/KT} + 2E N_0 e^{-2E/KT} \dots \infty E N_0 e^{-\infty E/KT}$$

$$\Rightarrow E = N_0 E \left[e^{-E/KT} + 2e^{-2E/KT} + \dots \infty e^{-\frac{\infty E}{KT}} \right]$$

$$\Rightarrow E = N_0 E e^{-E/KT} \left[1 + 2e^{-E/KT} \dots \infty e^{-(\infty-1)E/KT} \right]$$

$$\Rightarrow \underline{E = \frac{E N_0 e^{-E/KT}}{(1 - e^{-E/KT})^2}} \left[\because \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 \dots \right]$$

$$\Rightarrow \bar{E} = \frac{E}{N}$$

$$\Rightarrow \bar{E} = \frac{E N_0 e^{-E/KT} \times (1 - e^{-E/KT})}{(1 - e^{-E/KT})^2 \times N_0}$$

$$\Rightarrow \bar{E} = \frac{E e^{-E/KT}}{1 - e^{-E/KT}} = \frac{E}{\left(1 - \frac{1}{e^{E/KT}}\right)} \left(\frac{1}{e^{E/KT}}\right)$$

$$\Rightarrow \underline{\bar{E} = \frac{E}{e^{E/KT} - 1}}$$

$$\Rightarrow \bar{e} = \frac{h\nu}{e^{\frac{h\nu}{KT}} - 1}$$

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Also no of oscillators per unit volume are in range ν & $\nu + d\nu$ given by

$$N = \frac{8\pi\nu^2}{c^3} d\nu$$

Aug energy on multiplying by no of oscillator per unit volume in range ν & $\nu + d\nu$ will give Energy density $E_\nu d\nu$

$$\Rightarrow E_\nu d\nu = \frac{8\pi\nu^2}{c^3} d\nu \times \frac{h\nu}{e^{\frac{h\nu}{KT}} - 1}$$

$$\Rightarrow E_\nu = \frac{8\pi h\nu^3}{c^3} \times \frac{1}{e^{\frac{h\nu}{KT}} - 1}$$

$$\Rightarrow E_\nu d\nu = \frac{8\pi h\nu^3}{c^3} \times \frac{1}{e^{\frac{h\nu}{KT}} - 1} d\nu$$

$$\text{Also, } \nu = \frac{c}{\lambda} \Rightarrow |d\nu| = \frac{c}{\lambda^2} d\lambda$$

$$\Rightarrow E_\lambda d\lambda = 8\pi h \left(\frac{\nu}{c}\right)^3 \frac{1}{e^{\frac{h\nu}{KT}} - 1} \left(\frac{c}{\lambda^2}\right) d\lambda \quad \text{Put } \nu = \frac{c}{\lambda}$$

$$\Rightarrow E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda KT}} - 1} d\lambda$$

→ Proof for wein's law

$$E_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda KT} - 1} d\lambda$$

for wein's, shortest λ

$$\frac{hc}{\lambda KT} \leq 1$$

$$\Rightarrow \frac{1}{e^{hc/\lambda KT} - 1} \approx e^{-hc/\lambda KT}$$

$$E_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} e^{-hc/\lambda KT}$$

→ For Rayleigh Jeans law
larger λ

Open express
for e^x

$$e^{1/KT} = 1 + \frac{hc}{\lambda KT} + \frac{1}{2!} \left(\frac{hc}{\lambda KT} \right)^2 + \dots$$

Neglected

$$\Rightarrow e^{1/KT} = 1 + \frac{hc}{\lambda KT}$$

$$\Rightarrow E_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \times \frac{\lambda KT}{hc} d\lambda$$

$$\Rightarrow E_{\lambda} d\lambda = \frac{8\pi KT}{\lambda^4} d\lambda$$

→ Photoelectric effect

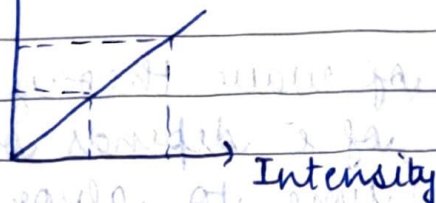
• work function → Min energy to eject e^-

→ FACTORS

• Effect of intensity

$V \rightarrow \text{constant} \leftarrow V$ $I \rightarrow \text{variable}$

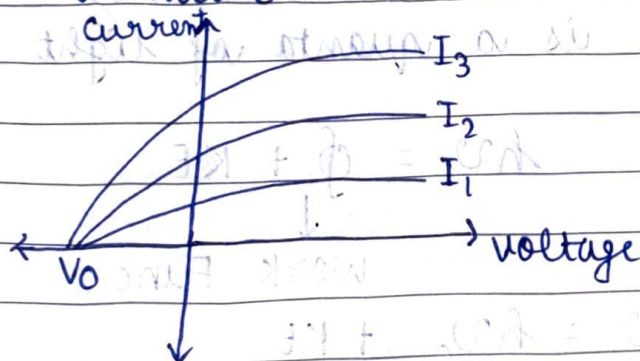
$\uparrow$ current



• effect of applied voltage

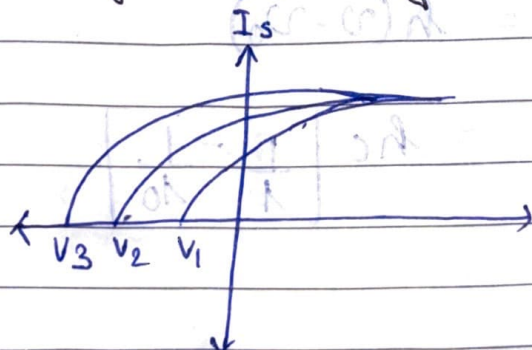
$V = \text{constant} = I$

$\uparrow$ $V \rightarrow \text{variable}$



$I_3 > I_2 > I_1$
→ V_0 doesn't
depend on
intensity

• effect of Frequency



$V_{03} > V_{02} > V_{01}$

→ No of photons emitted is independent of freq.

→ Energy of e^- depends on frequency

- Laws of photoelectric effect
- No. of photo e^- depend on intensity
 - e^- emit when energy $> \phi$
 - KE of e^- depend on freq. of light
 - KE independent of intensity
 - No time lag b/t incident ray & e^- emission

- Failures of wave theory
- emission of e^- depends on intensity
 - e^- takes time to absorb energy
 - $KE \propto$ intensity

- Einstein's Photoelectric Theory

Photon is a quanta of light having energy
 $E = h\nu$

$$h\nu = \phi + KE$$

↓
Work Func.

$$h\nu = h\nu_0 + KE$$

$$KE = h(\nu - \nu_0)$$

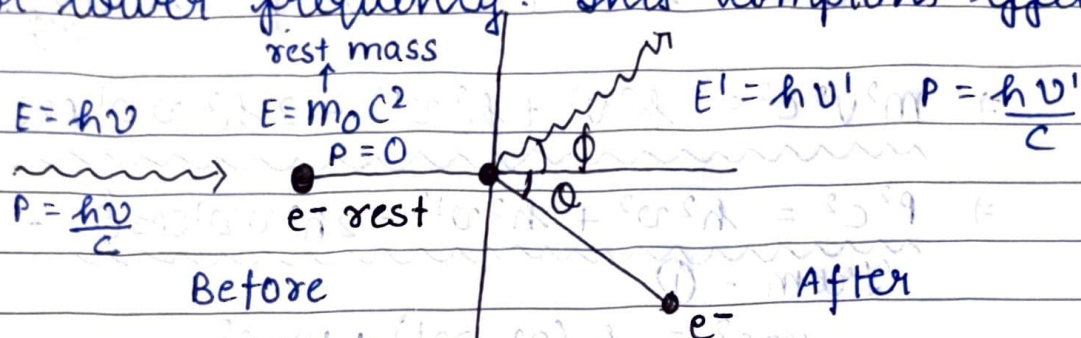
$$\frac{1}{2} m v_{\max}^2 = h(\nu - \nu_0)$$

$$\frac{1}{2} m v_{\max}^2 = hc \left[\frac{1}{\lambda} - \frac{1}{\lambda_0} \right]$$

→ $KE \propto \Delta\nu$

→ Intensity $\propto$ No. of photons

- Compton's effect
 when a monochromatic higher frequency or shorter wavelength electromagnetic wave is scattered then scattered wave has longer or lower frequency. This is Compton's effect



$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$ → rest mass

$E = mc^2, P = mv$

- Elastic collision (E & P conserved)
 $\Rightarrow h\nu + m_0 c^2 = mc^2 + h\nu' \quad \text{--- (1)}$
conserve energy

Along horizontal conserve P

$$\frac{h\nu}{c} + 0 = \frac{h\nu' \cos \phi}{c} + mv \cos \theta$$

$$\frac{h\nu}{c} = \frac{h\nu' \cos \phi}{c} + mv \cos \theta \quad \text{--- (2)}$$

Along vertical dirⁿ

$$0 = \frac{h\nu' \sin \phi}{c} - mv \sin \theta \quad \text{--- (3)}$$

$$\Rightarrow mv \sin \theta = \frac{h\nu' \sin \phi}{c} \quad \text{--- (4)}$$

$$\Rightarrow mv \cos \theta = \frac{h\nu}{c} - \frac{h\nu' \cos \phi}{c}$$

Squaring & Adding ③ & ④

$$m^2 v^2 c^2 (\cos^2 \theta + \sin^2 \theta) = (h\nu - h\nu' \cos \phi)^2 + (h\nu' \sin \phi)^2$$

$$\Rightarrow m^2 v^2 c^2 = h^2 \nu^2 + h^2 \nu'^2 \cos^2 \phi - 2h^2 \nu \nu' \cos \phi + h^2 \nu'^2 \sin^2 \phi$$

$$\Rightarrow m^2 v^2 c^2 = h^2 \nu^2 + h^2 \nu'^2 - 2h^2 \nu \nu' \cos \phi \quad - (5)$$

$$\Rightarrow p^2 c^2 = h^2 \nu^2 + h^2 \nu'^2 - 2h^2 \nu \nu' \cos \phi \quad - (6)$$

From - ①

$$mc^2 = h(\nu - \nu') + m_0 c^2$$

S.B.S

$$\Rightarrow m^2 c^4 = h^2 (\nu - \nu')^2 + m_0^2 c^4 + 2h m_0 c^2 (\nu - \nu')$$

$$\Rightarrow m^2 c^4 = h^2 (\nu^2 + \nu'^2 - 2\nu \nu') + m_0^2 c^4 + 2h m_0 c^2 (\nu - \nu')$$

$$\Rightarrow E^2 = h^2 (\nu^2 + \nu'^2 - 2\nu \nu') + m_0^2 c^4 + 2h m_0 c^2 (\nu - \nu') \quad - (7)$$

⑦ - ⑥

$$\Rightarrow E^2 - p^2 c^2 = 4 + 2h^2 \nu \nu' + 2h^2 \nu \nu' \cos \phi + m_0^2 c^4 + 2h m_0 c^2 (\nu - \nu')$$

Also we know $E^2 - p^2 c^2 = m_0^2 c^4$

$$\Rightarrow \cancel{2h^2 \nu \nu'} (\cos \phi - 1) + \cancel{2h m_0 c^2 (\nu - \nu')} = 0$$

$$h \nu \nu' (\cos \phi - 1) = m_0 c^2 (\nu' - \nu)$$

$$\Rightarrow \frac{h}{m_0 c^2} (\cos \phi - 1) = \frac{1}{\nu} - \frac{1}{\nu'}$$

UNIT-1 PAIR PRODUCTION



→ A photon of sufficiently high energy in strong electric field of nucleus give electron-positron pair

$$\Rightarrow \frac{1}{v_1} - \frac{1}{v} = \frac{h}{m_0 c^2} (1 - \cos \phi)$$

$$\Rightarrow \frac{c}{v_1} - \frac{c}{v} = \frac{h}{m_0 c} (1 - \cos \phi)$$

$$\Rightarrow \frac{1}{\lambda_1} - \frac{1}{\lambda} = \frac{h}{m_0 c} (1 - \cos \phi)$$

$$\Delta \lambda \propto \phi$$

→ UNIT-2

→ wave nature of Particle
de Broglie

→ Pair Production

→ It's a process in which a high energy photon is converted into a particle - antiparticle pair usually an electron e^- & positron (e^+) in presence of nucleus / another particle to conserve left momentum.

→ The photon must have energy at least equal to combined rest mass energy of particles.

$$E_\gamma \geq 2m_e c^2 \approx 1.022 \text{ MeV}$$

